Distributed Systems · Erlang/OTP · IoT/Telecom · Cloud Infra

Dat Nguyen

Principal Backend Architect — Erlang/OTP

Erlang/OTP engineer — Ericsson IMS, Cisco NSO automation, and distributed IoT backends.

Open to remote senior and principal roles

Contact GitHub (https://github.com/datnguyen-tt)

About

Principal Backend Architect and Erlang/OTP engineer with 10+ years building fault-tolerant distributed systems for IoT, telecom, network automation, and cloud-native environments.

Designs resilient services, leads cross-functional delivery, and diagnoses complex concurrency and production issues. Brings hands-on depth across OTP architecture, telecom protocols, observability, data stores, and container-based infrastructure.

- **10+** Years in backend engineering
- **3** Employers: 5GenCare, PxD, DEK
- **2** Client engagements: Cisco, Ericsson
- **1** IEEE ATC publication

Experience

5GenCare

offset: 4 2023-06 → present

Principal Backend Architect — Erlang/OTP Engineer

Hong Kong · Remote `IoT` `remote`

+ Architected and maintained a distributed IoT/AIoT backend with Erlang/OTP and Mnesia, supporting cloud telemetry and device management.
+ Designed fault-tolerant Erlang services and Prometheus monitoring for cloud telemetry workflows.
+ Led Docker deployments through GitLab CI for container-based backend releases.

`Erlang/OTP, Mnesia, Docker, GitLab CI, PostgreSQL, Prometheus, TLS/OpenSSL`

Precision Development (PxD)

offset: 3 2022-06 → 2023-06

Senior Full-Stack Engineer — Cloud Systems

Newton, Massachusetts · Remote `Telecom` `remote`

+ Built distributed voice and text services with Erlang and FreeSWITCH for low-connectivity environments.
+ Designed Flask and Cowboy APIs integrating USSD, IVR, SMS, WhatsApp, and Telegram communication flows.
+ Instrumented time-sensitive communication paths with metrics and tracing, then diagnosed production faults.

`Erlang, Python, Flask, Erlang Cowboy, FreeSWITCH, USSD/IVR/SMS, WhatsApp/Telegram APIs, Docker, AWS, PostgreSQL`

DEK Technologies Vietnam

offset: 2 2020-01 → 2022-06

Erlang Consultant for Cisco NSO (via DEK Technologies)

Ho Chi Minh City, Vietnam `Network automation` `remote`

+ Delivered Cisco NSO network automation services with Erlang, YANG, NETCONF, and RESTCONF.
+ Mapped YANG network models into maintainable domain logic using Domain-Driven Design.
+ Resolved concurrency defects in Erlang network automation services, strengthening fault tolerance.

`Erlang, Python, Cisco NSO, YANG, NETCONF, RESTCONF, Jenkins, Emacs Lisp, Bash`

DEK Technologies Sweden

offset: 1 2019-10 → 2019-12

Erlang Consultant for Cisco (via DEK Technologies)

Stockholm, Sweden `Network automation` `onsite`

+ Integrated Erlang and YANG services across sites with Cisco's architecture team.
+ Led OTP design and BEAM optimization sessions, transferring runtime knowledge to the engineering team.

`Erlang, YANG, C++, NETCONF, RESTCONF, Jenkins, Jira`

DEK Technologies Vietnam

offset: 0 2016-11 → 2019-11

Erlang Consultant for Ericsson IMS (via DEK Technologies)

Ho Chi Minh City, Vietnam Telecom onsite

+ Developed Ericsson IMS SBG components for media control and SIP, Diameter, and Megaco protocol handling.
+ Automated Jenkins CI pipelines for Erlang and Python telecom components, making build and test workflows repeatable.
+ Led testing across IMS protocol and media-control paths using Wireshark.

Erlang, Python, IMS, SIP, Diameter, Megaco, Jenkins, Wireshark

Skills

Core Erlang/OTP: Erlang/OTP, GenServer, Supervision Trees, Fault-tolerant OTP Design, State Machines, Hot Code Upgrade, BEAM VM, ETS/DETS

Distributed Systems & Architecture: Distributed Systems, Distributed Erlang, Actor Model, Domain-Driven Design, Cloud-native Microservices, Mission-critical Systems, High Availability, Fault Tolerance

Protocols & Telecom: SIP, Diameter, Megaco, IMS, NETCONF, RESTCONF, YANG, Cisco NSO

Data Stores: Mnesia, PostgreSQL, ETS/DETS, SQL, CiscoDB, NoSQL

Cloud & Observability: AWS, Docker, Docker Swarm, Prometheus, Grafana, GitLab CI, Jenkins, Wireshark

Programming Languages: Erlang/OTP, Python, Bash, YANG, C/C++, Emacs Lisp, JavaScript, YAML

Testing & Debugging: Common Test, Dialyzer, DBG, EDTS, Lux Test Framework, Tracing, Production Debugging

Projects

erl-trace **erlang_ts** **pingpong_application** **dshell**

prometheus-cowboy **edts**

Publications

2021 International Conference on Advanced Technologies for Communications (ATC) 2021

Integrated Multi-standard system based on LR1110 for geolocation applications

Bui Phung Huu Duc, Dat Tran Nguyen Tien, Phuong Binh Nguyen, Nguyen Minh Thien, Le Huy Trinh, Fabien Ferrero

DOI: 10.1109/ATC52653.2021.9598262

Education

DEGREES

2020-09 → 2022-05

Master of Science in Information Technology

University of Information Technology, Vietnam

- Built a hybrid geolocation gateway using LoRaWAN and Semtech LR1110.
- Co-authored an IEEE ATC 2021 paper on an LR1110-based integrated multi-standard geolocation system.
- Developed a LoRaWAN relay virtual gateway addressing installation, coverage, cost, and energy constraints.

2012-09 → 2016-06

Bachelor of Engineering in Computer Engineering

University of Information Technology, Vietnam

- Completed a final project on solar-powered LoRa sensor systems with predictive logic.
- Completed an internship on resume parsing with C++ and regular expressions.

AWARDS

2017 **First Prize — Student Scientific Research Projects**
University of Information Technology

2015 **Second Prize — Creative Ideas Grant**
University of Information Technology

2014 **Student of 5 Merits Award**
University of Information Technology

LANGUAGES

Vietnamese Native

English Upper-intermediate

Contact

Email datttn.work@gmail.com LinkedIn www.linkedin.com/in/datnguyen-t GitHub github.com/datnguyen-tt